Supporting Information

Rapid sequential touches complicate whisking-phase analysis in mouse active touch

Yuxuan Xu

Additional results

Throughout this supplement, an archive subject label means the subject label stored in the metadata of each Neurodata Without Borders file. Files sharing a label are combined. Because the identity mapping is unavailable, the label serves as an aggregation key without establishing a unique biological animal. For count models, the predictive score is 1 - Dmodel/Dnull, where D is mean Poisson deviance in trials omitted from fitting. A phase prediction value is the score after adding phase minus the score before adding it; 0.01 is a gain of one percentage point.

Supplemental Table S1. Touch spacing and mean spike count

Every touch in this analysis had an observed preceding onset and an observed following onset in the same trial. The same minimum interval was applied on both sides. The thresholds are nested, so each higher threshold retains a cumulative subset of the preceding condition. File means were first combined by their median for each archive subject label. The table gives the mean of those label values for spike counts during 0-50 ms after touch. Uncertainty is reported as a bootstrap 95% confidence interval (CI).

Minimum interval to both neighbors (ms)	Touches	Files	Archive subject labels	Mean spikes per touch	Bootstrap 95% CI
0	16,231	41	21	0.364	0.241 to 0.502
20	15,750	41	21	0.362	0.239 to 0.497
30	14,808	41	21	0.360	0.240 to 0.491
50	10,097	41	21	0.363	0.247 to 0.490
100	4,532	41	21	0.367	0.247 to 0.501
250	350	38	21	0.350	0.214 to 0.506

A complementary summary using nonoverlapping interval bands for the preceding and following intervals is included with the numerical source data.

Supplemental Table S2. Phase prediction across neighbor exclusion thresholds

The added fraction is the gain in predictive score when phase terms enter the model.

Neighbor threshold (ms)	Retained touches	Labels included	After timing and task	After adding movement
0	19,675	21	0.0124	0.0096
20	19,160	21	0.0124	0.0082
30	18,157	21	0.0134	0.0105
50	12,773	20	0.0152	0.0084
100	6,251	17	0.0086	0.0083
250	1,014	0	not estimated	not estimated

At 250 ms, no recording file met the requirements of at least 80 observations and at least 15 observations in each phase-defined half-cycle. The descriptive response remains available in Supplemental Table S1.

In the primary phase analysis, a missing neighbor at a trial boundary was treated as no recorded close onset. At the 50-ms threshold, a stricter analysis required an observed onset on each side. It selected 10,097 touches; 9,247 touches from 30 files and 19 archive subject labels met the phase model criteria. After timing and task variables, phase removed an additional median fraction of 0.0203 (95% CI, 0.0046 to 0.0336; $P < 0.001$). After measured movement was added, the value was 0.0079 (95% CI, 0.0008 to 0.0212; $P = 0.002$).

Supplemental Table S3. Baseline and response window sensitivity

The table crosses all response and baseline windows. One condition retained all preceding intervals; the comparison condition required a preceding interval of at least 100 ms. For each response width, touches with a following onset inside that window were excluded. Coefficients are standardized coefficients for the log preceding inter-touch interval. Q values control the false discovery rate across these 50 sensitivity estimates. The $q = 0.017$ in the manuscript comes from 30 combinations of six interval thresholds and five baseline widths. Five combinations at the 250-ms threshold contained fewer than five archive subject labels, so 25 comparisons entered that adjustment. Requiring five labels avoids group tests based on very small sets; smaller sets remain descriptive. The $q = 0.010$ reported here comes from a separate adjustment across these 50 response and baseline combinations. Complete numerical results, sample counts, and confidence intervals are included with the numerical source data.

Selected-event counts include every touch that met the spacing and window rules. History coefficient summaries include files with at least 80 complete observations. Phase summaries additionally require at least 15 observations in each phase-defined half-cycle and four trials. For count phase models, a file and response-window combination is omitted if any fitting fold contains no spikes. The corresponding modeled-event counts are reported in the manuscript captions and numerical source data.

Response (ms)	Baseline (ms)	All intervals coefficient	All intervals q	100-ms coefficient	100-ms q
10	10	0.037	0.035	0.014	0.844
10	20	0.077	0.010	0.010	0.751
10	30	0.085	0.009	0.012	0.751
10	40	0.084	0.009	0.012	0.751
10	50	0.080	0.009	0.015	0.603
20	10	0.088	0.010	0.053	0.350
20	20	0.098	0.009	0.053	0.228
20	30	0.110	0.009	0.030	0.291
20	40	0.120	0.009	0.020	0.279
20	50	0.117	0.009	0.046	0.200
30	10	0.097	0.011	0.027	0.751
30	20	0.108	0.010	0.011	0.751
30	30	0.125	0.009	0.008	0.822
30	40	0.121	0.009	0.002	0.797
30	50	0.116	0.010	0.016	0.751
40	10	0.079	0.016	-0.006	0.813
40	20	0.116	0.015	-0.017	0.895
40	30	0.118	0.011	-0.008	0.895
40	40	0.114	0.012	-0.007	0.941
40	50	0.106	0.016	-0.015	0.895
50	10	0.095	0.020	-0.015	0.751
50	20	0.123	0.011	-0.015	0.895
50	30	0.161	0.010	-0.013	0.895
50	40	0.134	0.010	-0.004	0.895
50	50	0.132	0.010	-0.017	1.000

Phase prediction across response windows

The median additional fractions removed after timing and task variables were 0.0035, 0.0044, 0.0127, 0.0121, and 0.0152 for 10-, 20-, 30-, 40-, and 50-ms response windows. After measured movement was added, the corresponding values were 0.0030, 0.0036, 0.0073, 0.0077, and 0.0084.

Interaction between amplitude and phase

This analysis tested whether the cyclic phase profile changed continuously with whisking amplitude. It added interactions between log amplitude and each of the four phase terms after the phase and measured movement variables. The selected set contained 12,773 touches; 12,296 touches from 35 files and 20 archive subject labels met the model criteria. The median additional fraction of Poisson deviance removed was 0.0025 (14/20 labels positive; bootstrap 95% CI, -0.0002 to 0.0150; Wilcoxon P = 0.030; sign test P = 0.115). The small median increase and interval spanning zero indicate limited and uneven evidence that the phase profile varied with amplitude.

Supplemental Table S4. Effect of phase representation and regularization

The first column lists variables included before phase terms were added. The representation with four phase terms uses sin(theta), cos(theta), sin(2theta), and cos(2theta). The representation with the first pair uses only sin(theta) and cos(theta). Following the archived sign convention, the separate protraction and retraction model represents progress from 0 to 1 within each phase-defined half-cycle and fits an independent quadratic curve for each half. Instantaneous direction came from the velocity measure. The added fraction has the same meaning as in Supplemental Table S2.

Existing variables	Phase representation	L2 regularization strength	Median added fraction	Positive labels	Wilcoxon P	Sign test P
Timing and task	Four phase terms	0.01	0.0123	13/20	0.154	0.263
Timing and task	Four phase terms	0.1	0.0137	12/20	0.036	0.503
Timing and task	Four phase terms	1	0.0152	15/20	<0.001	0.041
Timing and task	Four phase terms	10	0.0054	15/20	<0.001	0.041
Timing and task	First sine and cosine pair	0.1	0.0094	12/20	0.064	0.503
Timing and task	First sine and cosine pair	1	0.0097	15/20	0.002	0.041
Timing and task	First sine and cosine pair	10	0.0022	15/20	<0.001	0.041
Timing and task	Separate protraction and retraction curves	0.1	0.0081	13/20	0.053	0.263
Timing and task	Separate protraction and retraction curves	1	0.0055	16/20	<0.001	0.012
Timing and task	Separate protraction and retraction curves	10	0.0028	16/20	<0.001	0.012
Timing, task, movement	Four phase terms	0.01	0.0070	13/20	0.189	0.263
Timing, task, movement	Four phase terms	0.1	0.0137	12/20	0.064	0.503
Timing, task, movement	Four phase terms	1	0.0084	15/20	0.002	0.041
Timing, task, movement	Four phase terms	10	0.0050	15/20	<0.001	0.041
Timing, task, movement	First sine and cosine pair	0.1	0.0014	10/20	0.202	1.000
Timing, task, movement	First sine and cosine pair	1	0.0040	13/20	0.017	0.263
Timing, task, movement	First sine and cosine pair	10	0.0018	15/20	<0.001	0.041
Timing, task, movement	Separate protraction and retraction curves	0.1	-0.0007	10/20	0.648	1.000

Existing variables	Phase representation	L2 regularization strength	Median added fraction	Positive labels	Wilcoxon P	Sign test P
Timing, task, movement	Separate protraction and retraction curves	1	0.0027	14/20	0.027	0.115
Timing, task, movement	Separate protraction and retraction curves	10	0.0020	16/20	<0.001	0.012

The reference regularization strength was 1.0, and no hyperparameter tuning was performed. With four phase terms, the median added fraction was positive at every regularization strength. Wilcoxon and sign test results varied across settings, showing that the estimated phase contribution depends on the chosen phase terms and regularization strength.

Spike-occurrence sensitivity

Logistic models predicted whether any spike occurred during 0-50 ms after touch. Their predictive score used binary log loss in place of Poisson deviance. After timing and task variables, phase increased the score by a median 0.0103 (12/20 labels positive; bootstrap 95% CI, -0.0023 to 0.0536; Wilcoxon q = 0.177; sign-test q = 0.503). After movement variables were added, the median gain was 0.0097 (12/20 positive; 95% CI, -0.0128 to 0.0390; Wilcoxon q = 0.349; sign-test q = 0.503). These two comparisons and the two reference count comparisons formed the four-test phase family.

Archive label merge sensitivity

Each of the 210 possible pairs of archive subject labels was merged in turn. For the dense baseline-subtracted analysis, the median coefficient ranged from 0.125 to 0.140 and Wilcoxon q ranged from 0.008 to 0.032. At the 100-ms preceding interval, the median ranged from -0.023 to 0.004 and q ranged from 0.702 to 1.000. For the count phase models, median gains remained positive. Before movement adjustment, Wilcoxon q ranged from <0.001 to 0.003 and sign-test q from 0.038 to 0.127. After movement adjustment, Wilcoxon q ranged from 0.002 to 0.008 and sign-test q from 0.038 to 0.127. This analysis addresses one duplicated identity at a time.

Supplemental Table S5. Continuous movement and contact coefficients

For each measure, file coefficients were combined by their median for each archive subject label, using touches without a recorded neighboring onset within 50 ms. Q values control the false discovery rate across the eight coefficients. Signed and absolute peak curvature changes use the transformations defined in Methods. Values for individual archive subject labels are supplied in the numerical source data.

Measure	Archive subject labels	Median standardized coefficient	Bootstrap 95% CI	q
Whisking amplitude	20	0.084	0.009 to 0.113	0.051
Angular velocity before touch	20	0.041	-0.013 to 0.112	0.418
Angular acceleration before touch	20	0.022	-0.002 to 0.056	0.118
Signed peak curvature change	20	0.031	-0.034 to 0.088	0.603
Absolute peak curvature change	20	0.030	-0.047 to 0.126	0.418

Measure	Archive subject labels	Median standardized coefficient	Bootstrap 95% CI	q
Touch duration	20	0.025	-0.038 to 0.097	0.418
Distance to pole	20	-0.005	-0.035 to 0.026	0.869
Base angle at touch	20	-0.104	-0.201 to 0.118	0.869

Contact duration across the phase-defined half-cycles

The primary phase set contained 10,113 touches in the protraction half-cycle and 2,660 in the retraction half-cycle, each with a paired contact duration. Median contact duration was 18 ms in the protraction half-cycle and 44 ms in the retraction half-cycle. Contacts lasting at least 50 ms accounted for 17.1% and 45.3% of touches in the respective half-cycles. The neighbor onset rule removes closely timed new contacts and permits the current contact to continue through part of the response window.

Direct comparison of the phase-defined half-cycles

The direct retraction-half minus protraction-half difference in raw spikes per touch was -0.043 (20 archive subject labels; 95% CI, -0.141 to 0.026; $P = 0.202$). After timing and task variables were included, the difference was -0.117 standardized units (95% CI, -0.227 to 0.040; $P = 0.143$). After measured movement was added, it was -0.030 (95% CI, -0.169 to 0.047; $P = 0.245$). These estimates compare weighted means for each phase-defined half-cycle within each recording file, followed by one median per archive subject label.

Contact measures after onset

Signed and absolute curvature change and contact duration were added to the prediction model. The four sine and cosine phase terms then removed an additional median fraction of 0.0038 (20 archive subject labels; 12 positive; 95% CI, -0.0022 to 0.0139; $P = 0.070$). These variables overlap the 50-ms response interval. The result is therefore descriptive.

Recording location sensitivity

In the primary model that included measured movement, phase removed an additional median fraction of 0.0210 in 25 C2 files from 12 archive subject labels (95% CI, 0.0084 to 0.0348). The value was 0.0017 in 10 files from nearby barrels and 9 labels (95% CI, -0.0035 to 0.0075). One label contributed files to both groups, so the estimates are descriptive subgroup summaries.